Lab 2 Frequently Asked Questions

Materials and Equipment

What size paper will we use?
The paper will be standard US letter size: 8.5 × 11 inches.

What are the dimensions of the ink pad?
I need to measure this. For the moment, assume a large one, on the order of 10 x 6cm

Will every group have its own stamp setup?
No. Groups will need to share the available stamp setups. The total number of setups will be confirmed shortly.

Do we need to buy a camera?
No. Use a camera that you already have easy access to, such as a cell phone camera, webcam, or laptop camera.

How many cameras may we use?
A specific limit has not yet been provided. The lab is designed so that it can be completed using readily available camera equipment.

Camera Placement

Is there a camera mounted on the crane?
No. The crane cannot support the weight of a camera.

May we attach a camera to the crane's end effector?
No. The crane and its end effector are not designed to carry a camera.

May we build a separate camera mount that provides an overhead view of the plate?
Yes. You may create a camera mount that provides an overhead view of the work area.

CAD Models and Dimensions

Is there a CAD model of the stamp holder and stamp setup?
Yes. The Onshape document contains the relevant geometry, including the information needed to determine the offset between the AprilTag center and the disc center in the stamp's coordinate frame.

In reality, I'm currently working through a permissions problem. I'll update this soon.

What does “plate” mean in the instructions?
The “plate” is the overall base plate of the experimental setup.

What are the dimensions of the plate?
The plate dimensions should be obtained from the shared Onshape model once the access issue has been resolved.

How are the stamps positioned relative to one another?
The positions of the stamps are defined in the Onshape model.

AprilTags and Stamp Placement

Are AprilTags attached to the stamp holders?
Yes. AprilTags will be placed at the base of the stamp holder.

Are AprilTags attached to the stamps themselves?
No. The stamps are too small to accommodate AprilTags large enough to be detected reliably.

How should the robot return a stamp if the stamp itself does not have an AprilTag?
Return the stamp to the same location from which it was picked up. Your pickup routine should be precise and repeatable enough to return the stamp to that location. You may also use camera feedback to improve the positioning accuracy of the end effector.

Lab Instructions and Course Materials

The image-processing pipeline in the written instructions differs from the pipeline shown in class. Which one should we follow?
You may develop your own pipeline. The provided pipeline is a suggested way to approach the problem, not a required implementation. The graphics in the instructions and slides will be updated so that they are consistent.